

# PARAM KAMLESHKUMAR PATEL

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## Education

### Stony Brook University

MS in Computer Science (GPA: 3.67/4.00)

August 2024 – May 2026 (Expected)

Stony Brook, NY, USA

### Indus University

B.Tech in Computer Engineering (GPA: 9.9/10.0)

August 2020 – May 2024

Ahmedabad, GJ, India

## Technical Skills

**Languages(Certified):** Python, Java, C, C++, JavaScript, SQL/PLSQL

**Frameworks:** PyTorch, TensorFlow, Scikit-Learn, NodeJS, NextJS, ReactJS, AngularJS, Spring, FastAPI, Quart

**Concepts:** OOPs, REST API, MVC, Generative AI, LLMs, LangChain, Web Scraping, Automation, OCI, A2A, MCP

**Database:** MongoDB, PostgreSQL, Neo4j, Vector Databases (Pinecone, Faiss, Chroma)

**Tools:** Git, Docker, PowerBI, Linux, Tableau, LangSmith, Eclipse, Splunk, Dynatrace, Outlook, Postman, Android Studio

## Experience

### Arch Mortgage Insurance (Arch MI)

June 2025 – August 2025

Generative AI Intern

Greensboro, NC, USA

- Designed a **multi-agent** automation platform coordinating Splunk, Dynatrace, Veracode, GitHub & Outlook to support self-healing IT workflows
- Built supervisor-agent **orchestration** using LLM-driven reasoning and structured output validation
- Automated** PR generation and CI/CD deployment workflows aimed at cutting incident triage time by ~ 75%

### Uminber Designs

January 2024 – April 2024

AI Intern

Ahmedabad, GJ, India

- Released a **production RAG** chatbot using OpenAI models + LangChain, increasing user task completion by 31%
- Improved answer accuracy 25% via custom **prompt-routing**, **chain refinement**, and **synthetic QA generation**
- Deployed full-stack (Next.js + PostgreSQL) with **Dockerized** inference achieving 99% uptime under load

## Projects

### HeartBridgeAI – AI Emotional Support Platform

Param2k23/HackPrinceton

- Developed **multi-agent system** using Grok AI, ElevenLabs, and Gemini to analyze mood from parent-child conversations with 95%+ intent-detection accuracy
- Engineered event-recommendation engine with **Amazon Nova**, scheduling stress-reducing activities via Google Calendar
- Delivered React web app with mood dashboards, family-memory pages, and secure PostgreSQL storage supporting 1,000+ interactions/day

### Multi-Model Debate for Efficient Retrieval

Param2k23/NLP Project

- Constructed a biomedical **QA system** combining RAG retrieval, lightweight LLM generators, and RoBERTa-based classification across 1k+ PubMedQA samples
- Implemented a debate-driven explanation module using Flan-T5-Large and Qwen3-0.6B to refine reasoning
- Designed a **vector retrieval** workflow that embeds all PubMed contexts and surfaces top-5 evidence passages
- Fine-tuned the retriever to improve **context relevance**, boosting the quality of downstream LLM-generated explanations

### Humor Understanding in Large Language Models

Param2k23/DSF Project

- Fine-tuned** RoBERTa and GPT-2 on Reddit/Twitter datasets to classify humor on 120k Reddit/Twitter samples, achieving 74% accuracy
- Developed preprocessing pipelines with **token filtering**, **semantic similarity scoring**, and **transformer-based** embeddings cutting noise by ~ 18%
- Benchmarked model outputs against Claude 3.5 and curated **human-labeled** sets to evaluate reasoning quality and labeling consistency

### Social Media Analysis of the U.S. Presidential Election

Document

- Processed 40M+ tweets using Spark with VADER and NRC lexicons to analyze major **sentiment** spikes
- Built a DBSCAN-based **bot-detection** pipeline that improved overall sentiment-signal clarity by 22%

## Achievements

- Earned the **Oracle OCI Generative AI Professional** certification, demonstrating advanced expertise and professionalism in cloud-based generative AI technologies, accompanied by an official credential badge
- Won 3** hackathons including HackNYU, Hopper Hacks, and Arch MI by delivering competitive technical solutions